

## SLAVGOROD, Russian Federation

WMO: 299150

Lat: 52.967N

Lon: 78.650E

Elev: 126

StdP: 99.82

Time Zone: 7.00 (E07)

Period: 94-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-35.3 | (c)<br>-32.0 | (d)<br>-38.2                  | (e)<br>0.1 | (f)<br>-35.2 | (g)<br>-34.8 | (h)<br>0.1 | (i)<br>-32.0 | (j)<br>11.0           | (k)<br>-4.0 | (l)<br>9.4 | (m)<br>-6.8 | (n)<br>1.9            | (o)<br>340 | (p)<br>0.610 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.5                   | 32.2            | 18.1 | 30.4 | 18.0 | 28.6 | 17.5 | 20.4                | 27.5 | 19.5 | 26.6 | 18.7 | 25.8 | 3.9                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.1                            | 13.2 | 22.6 | 17.1 | 12.4 | 22.0 | 16.2 | 11.7 | 21.4 | 59.3          | 27.6 | 56.4 | 26.4 | 53.7 | 25.9 | 26.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 9.0           | (b) 7.8 | (c) 7.0 | (e) -36.8                  | (f) 35.6 | (g) 4.1            | (h) 2.2 | (i) -39.7                                          | (j) 37.2 | (k) -42.1  | (l) 38.5 | (m) -44.4  | (n) 39.7 | (o) -47.3  | (p) 41.3 |    |    |
| (a) 9.0           | (b) 7.8 | (c) 7.0 | (e) -36.9                  | (f) 21.9 | (g) 4.0            | (h) 1.7 | (i) -39.8                                          | (j) 23.1 | (k) -42.2  | (l) 24.0 | (m) -44.4  | (n) 25.0 | (o) -47.3  | (p) 26.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 3.1    | -18.0 | -15.4 | -7.0  | 5.8   | 13.5  | 19.6  | 20.9  | 18.7  | 12.2  | 4.4   | -5.6  | -13.5 |
|                                                                                  | DBStd       | 15.02  | 8.24  | 7.09  | 6.82  | 5.88  | 5.37  | 4.25  | 3.49  | 3.91  | 4.84  | 5.15  | 7.69  | 7.73  |
|                                                                                  | HDD10.0     | 3701   | 868   | 712   | 527   | 150   | 28    | 1     | 0     | 1     | 31    | 186   | 469   | 729   |
|                                                                                  | HDD18.3     | 5824   | 1127  | 945   | 785   | 376   | 167   | 36    | 15    | 44    | 192   | 433   | 719   | 987   |
|                                                                                  | CDD10.0     | 1163   | 0     | 0     | 0     | 25    | 136   | 288   | 337   | 270   | 97    | 11    | 0     | 0     |
|                                                                                  | CDD18.3     | 246    | 0     | 0     | 0     | 1     | 16    | 73    | 93    | 54    | 8     | 0     | 0     | 0     |
|                                                                                  | CDH23.3     | 2557   | 0     | 0     | 0     | 17    | 232   | 753   | 853   | 584   | 117   | 1     | 0     | 0     |
|                                                                                  | CDH26.7     | 858    | 0     | 0     | 0     | 3     | 67    | 264   | 294   | 192   | 38    | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.4    | 3.4   | 3.5   | 3.7   | 3.8   | 3.7   | 3.2   | 2.9   | 3.0   | 3.2   | 3.4   | 3.7   | 3.8   |
| Precipitation                                                                    | PrecAvg     | 297    | 15    | 13    | 15    | 16    | 22    | 39    | 55    | 34    | 22    | 25    | 22    | 19    |
|                                                                                  | PrecMax     | 433    | 34    | 30    | 42    | 38    | 80    | 127   | 152   | 89    | 46    | 67    | 55    | 33    |
|                                                                                  | PrecMin     | 174    | 2     | 1     | 2     | 1     | 3     | 10    | 6     | 3     | 5     | 4     | 9     | 6     |
|                                                                                  | PrecStd     | 71     | 8     | 7     | 9     | 10    | 16    | 30    | 33    | 18    | 11    | 17    | 10    | 7     |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 1.3   | 1.0   | 9.7   | 25.7  | 32.1  | 33.9  | 34.2  | 34.6  | 31.3  | 21.1  | 10.2  | 1.7   |
|                                                                                  |             | MCWB   | 0.6   | 0.2   | 5.6   | 13.9  | 16.7  | 18.4  | 19.6  | 18.8  | 16.9  | 11.7  | 6.7   | 0.6   |
|                                                                                  | 2%          | DB     | -1.3  | -0.8  | 5.5   | 21.7  | 28.4  | 31.6  | 31.7  | 31.1  | 26.5  | 17.6  | 7.4   | 0.3   |
|                                                                                  |             | MCWB   | -2.1  | -1.5  | 3.2   | 12.4  | 15.3  | 17.5  | 19.2  | 18.2  | 14.9  | 10.3  | 5.0   | -0.6  |
|                                                                                  | 5%          | DB     | -3.5  | -2.7  | 2.9   | 18.6  | 26.0  | 29.7  | 29.9  | 28.6  | 23.4  | 14.5  | 5.0   | -1.3  |
|                                                                                  |             | MCWB   | -4.2  | -3.4  | 1.4   | 10.3  | 14.5  | 17.4  | 18.8  | 17.7  | 13.5  | 8.9   | 3.2   | -2.0  |
|                                                                                  | 10%         | DB     | -5.6  | -4.8  | 1.5   | 15.6  | 23.4  | 27.7  | 28.0  | 26.6  | 20.5  | 11.8  | 2.9   | -3.1  |
|                                                                                  |             | MCWB   | -6.3  | -5.3  | 0.4   | 9.0   | 13.4  | 16.9  | 18.3  | 16.8  | 12.4  | 7.5   | 1.7   | -3.8  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 0.8   | 0.2   | 6.1   | 14.7  | 18.0  | 20.4  | 21.7  | 20.9  | 17.8  | 12.6  | 7.7   | 0.9   |
|                                                                                  |             | MCDB   | 1.2   | 0.7   | 9.2   | 23.3  | 28.4  | 29.0  | 28.8  | 27.7  | 28.9  | 18.6  | 9.4   | 1.3   |
|                                                                                  | 2%          | WB     | -1.9  | -1.5  | 3.4   | 12.7  | 16.4  | 19.3  | 20.7  | 19.5  | 15.8  | 10.9  | 5.1   | -0.4  |
|                                                                                  |             | MCDB   | -1.3  | -0.9  | 5.4   | 20.6  | 25.9  | 27.1  | 27.8  | 27.6  | 23.6  | 16.4  | 7.1   | 0.2   |
|                                                                                  | 5%          | WB     | -4.1  | -3.3  | 1.7   | 11.0  | 15.3  | 18.5  | 20.0  | 18.6  | 14.4  | 9.4   | 3.4   | -2.0  |
|                                                                                  |             | MCDB   | -3.5  | -2.7  | 2.8   | 17.4  | 23.8  | 26.1  | 26.6  | 26.6  | 21.9  | 13.8  | 5.0   | -1.3  |
|                                                                                  | 10%         | WB     | -6.3  | -5.4  | 0.5   | 9.4   | 14.0  | 17.8  | 19.2  | 17.7  | 13.0  | 7.8   | 1.7   | -3.8  |
|                                                                                  |             | MCDB   | -5.7  | -4.8  | 1.4   | 15.0  | 21.9  | 25.2  | 25.6  | 25.0  | 19.3  | 11.5  | 2.9   | -3.1  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.2   | 8.7   | 8.6   | 11.1  | 12.3  | 11.2  | 10.5  | 11.4  | 11.1  | 8.8   | 6.8   | 8.0   |
|                                                                                  |             | MCDBR  | 9.4   | 8.2   | 8.2   | 15.7  | 15.8  | 14.2  | 12.9  | 14.7  | 15.7  | 13.6  | 7.8   | 9.1   |
|                                                                                  |             | MCWBR  | 9.2   | 8.0   | 6.5   | 8.0   | 6.6   | 4.8   | 4.0   | 5.0   | 6.5   | 7.3   | 5.8   | 9.0   |
|                                                                                  | 5% WB       | MCDBR  | 9.5   | 8.4   | 7.8   | 14.5  | 13.6  | 11.6  | 10.6  | 12.5  | 13.9  | 12.3  | 7.4   | 9.1   |
|                                                                                  |             | MCWBR  | 9.3   | 8.1   | 6.4   | 8.0   | 6.6   | 4.7   | 4.0   | 4.9   | 6.3   | 7.3   | 5.7   | 9.0   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.298  | 0.331 | 0.386 | 0.407 | 0.410 | 0.414 | 0.440 | 0.420 | 0.375 | 0.351 | 0.334 | 0.305 |       |
|                                                                                  | taud        | 2.026  | 1.963 | 1.927 | 2.153 | 2.196 | 2.271 | 2.201 | 2.223 | 2.296 | 2.260 | 2.113 | 1.995 |       |
|                                                                                  | Ebn at Noon | 636    | 732   | 771   | 820   | 839   | 838   | 806   | 799   | 793   | 722   | 589   | 535   |       |
|                                                                                  | Edh at Noon | 78     | 114   | 147   | 134   | 135   | 127   | 134   | 124   | 102   | 85    | 72    | 67    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.16   | 2.27  | 3.74  | 4.75  | 5.92  | 6.32  | 5.71  | 4.95  | 3.56  | 1.92  | 0.95  | 0.78  |       |
|                                                                                  | RadStd      | 0.10   | 0.11  | 0.30  | 0.34  | 0.46  | 0.42  | 0.38  | 0.34  | 0.30  | 0.25  | 0.11  | 0.11  |       |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (12 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | -0.46 | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page